

ABSTRACT

On July of 2020, my father helped me buy a 2011 Subaru Impreza with 64k miles for \$6k. On December 6th, 2021, I got in a car accident where my insurance deemed my car a total loss. My Impreza had 87k miles (23k more miles) & I received \$7.2k from my insurance (\$1.2k more than what I paid for it). To give you a look at the current used car market: If I was to buy the same Impreza, I bought **2 years ago** with the same mileage, it would cost me \$8.8-\$11.5k (that’s a **46%-91% price increase**).

Now that I’m carless again, I decided that this time, I would look for a car **myself**. Not knowing anything about cars, I decided data science would be useful tool. Having a budget of 15k, data science is going to help tailor my search.

INTRODUCTION

Using Kaggle I acquired a dataset called ‘100k UK Used Car Data set’ to conduct this analysis. From this data set, I focused on 6 variables: Engine Size, Year, Mileage, Transmission, Make, and Fuel Type. These variables will have an overlay of my budget to help visualize what is associated with a car costing more than \$15k.

REFERENCES

“2011 Subaru Impreza Values & Cars For Sale | Kelley Blue Book.
<https://www.kbb.com/subaru/impreza/2011/>.
Aditya. “100,000 UK Used Car Data Set.” *Kaggle*, 4 July 2020,
<https://www.kaggle.com/adityadesai13/used-car-dataset-ford-and-mercedes>.

EXPLORATORY DATA ANALYSIS

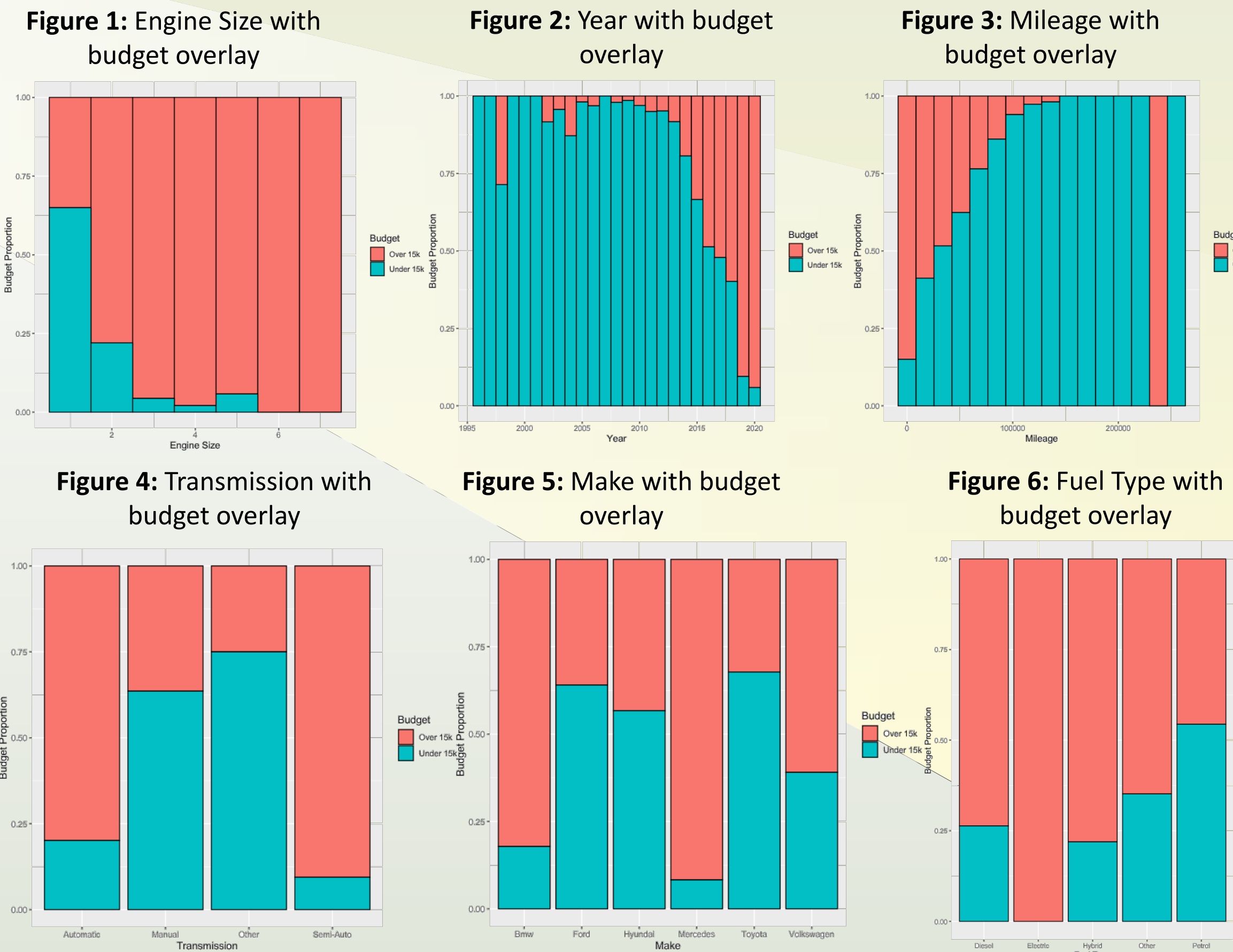


Figure 1 shows that as engine size increases, the cost of a car increases as well.

Figure 2 shows that newer cars are more likely to cost more than \$15K. 90% of cars built in 2019-20 cost more than \$15k.

Figure 3 shows that as the mileage of a car increases, it is more likely to cost less than \$15k.

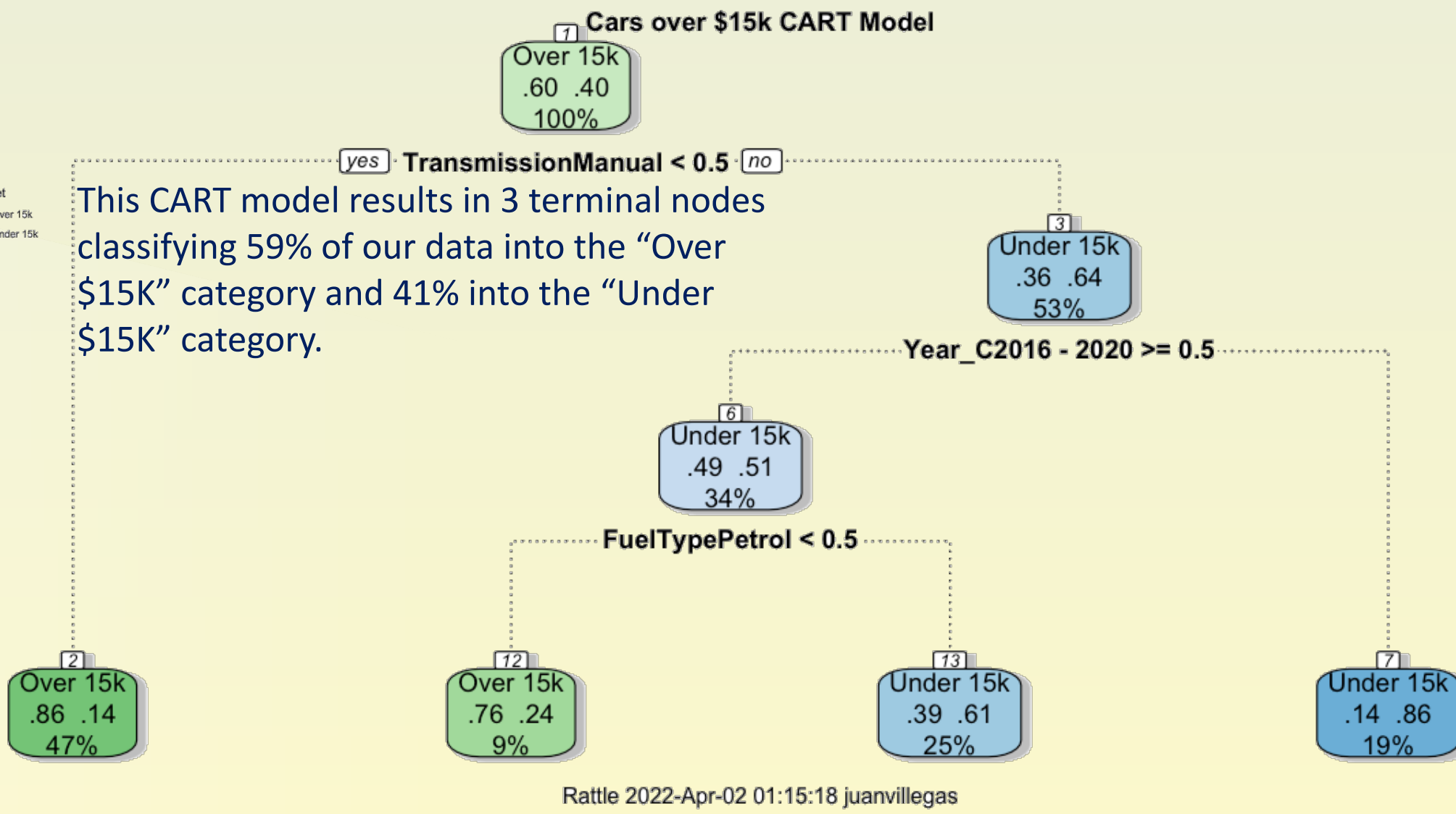
Figure 4 Shows that Semi-Auto & Automatic cars are more expensive than Manual or *Other (Other refers to **Continuously variable transmission (CVT)**) vehicles.

Figure 5 shows that Toyota, Fords & Mercedes are the brands with the highest proportion of cars costing less than \$15K. They all have a proportion higher than .58.

Figure 6 shows that most the Fuel Type associated most with car costing less is Petrol. All Electric Vehicles are more than \$15k.

Model & Results

METHODOLOGY
To obtain our results, 10-fold cross validation was used on CART decision tree models.



The best way to find a car for less than \$15k is to look for a car that has a manual Transmission and was made before 2016. If you're looking for a car made after 2016, make sure it runs on Petrol.

CONCLUSION

Our model shows the most strength when it identifies cars that cost more than our budget of \$15,000. It correctly Identifies these cars 83.44% of the time. Additionally, this model has a high specificity of 83% when identifying cars over \$15,000. This is useful to know because when deciding to buy a car, I want accurate information on what makes a car expensive so I can avoid these things. This saves me time by giving me a narrow list of attributes to look for in cars.

The Overall accuracy of our entire model is 80%, which is good considering that 60% of the cars in this entire data set are over our budget.

It is apparent that 2 attributes that impact a cars high price is: The type of Transmission and the Year. This is going to reduce the number of cars I look for when I search for a used car!